

Infrastructure Assessment & Scaling Strategy v2

Upgrading Data Infrastructure for Growth & Location-Based Pricing

Executive Summary

The EasiShop grocery price comparison platform currently operates on a single laptop for data collection and minimal cloud hosting with Render. While this has allowed us to validate the concept with four retailers (Pick n Pay, Woolworths, Shoprite and Checkers), collecting data from their Cape Town locations, the infrastructure presents critical risks and limits our ability to scale. This report outlines:

- Current infrastructure vulnerabilities
- Manual operational bottlenecks
- Three scalable architecture paths (on-premise servers, cloud-native, hybrid)
- Database migration options with pricing
- Automation opportunities using AI agents
- Recommended 6-month roadmap

Section 1: Current State Assessment

1.1 Data Collection Hardware

Component	Current Spec	Constraint
Device	HP ProBook 640 G2 laptop (4GB RAM, Intel i5)	Not designed for 24/7 operation
Concurrent jobs	2 scraping processes	Only 2 retailers at a time
Geographic coverage	Cape Town only	Cannot offer location-based pricing
Update frequency	1 - 2 times daily	Stale data, poor user experience

Reliability: Critical Risk

Single point of failure. If this laptop crashes, overheats, or loses internet, all data collection stops with no automatic recovery.

1.2 Platform Hosting (Render)

Plan	CPU	RAM	Monthly Cost
Starter (current)	0.5 core	512 MB	\$7 (R~130)

Pain Point: Long-running requests (e.g., complex price comparisons) time out due to insufficient compute. Users see errors instead of results.

Reliability: Low Risk - Render maintains a strong infrastructure meaning reliable hosting.

1.3 Storage & Authentication

Component	Current Solution	Risk
Product & price data	SQLite (single file)	File corruption = total data loss. No backups. Not designed for concurrent writes.
User accounts	PocketBase	Adequate for current scale, but not tested beyond hundreds of users

Security Risk: SQLite file is unencrypted and accessible if the laptop is compromised.

Section 2: Daily Operations & Automation Opportunities

2.1 Current Manual Process (Daily)

Every day, someone must manually execute the following sequence:

1. SSH into laptop or physically access it
2. Stop the running scraper processes
3. Save collected data to SQLite
4. Copy .db file to the Render-hosted application (manual upload or SCP)
5. Restart the scraper
6. Verify data appears correctly in user-facing app
7. Deploy User-facing app

Estimated daily effort: 20 to 30 minutes each update

Risk of human error: High (No updates if Admin is Out of Commission, forgetting steps, corrupting the db file, leaving scraper off)

2.2 Proposed Automation Using AI Agents & Scripting

Overtime, we would like to replace the manual pipeline with an event-driven automation workflow:

Task	Current Method	Proposed Automation	Tool Options	Estimated Monthly Cost
Stop/start scrapers	Manual SSH	Cron-scheduled scripts	Cron + Bash	\$0
Database export & transfer	Manual copy	Automated ETL pipeline	Airflow (free), Prefect (free tier)	\$0-50 (~835)
Data validation	Visual check	Automated checksums + anomaly detection	Python scripts + basic ML	\$0
Error recovery	Manual reboot	Watchdog daemon + auto-restart	Monit, systemd	\$0
AI Agent for scraping logic updates	N/A	AI agent adjusts selectors when retailer websites change	Browse.ai, BrightData AutoScraper, or custom LangChain agent	\$100-\$500 (~R1,680 – R8,360)

AI Agent Details: A LangChain-based agent could monitor scraper failures, detect when retailer HTML changes, and update XPath/CSS selectors automatically. This reduces maintenance from hours per week to near-zero.

Total automation investment: ~ \$150-\$600 (R2,500 - R10,300)/month depending on AI agent sophistication. This removes the human single point of failure and allows us to focus on more high leveraging activities.

Section 3: Scaling Options - Two Architectures

Instead of only proposing refurbished servers (which still require physical management), we present two viable paths.

Option A: Refurbished Virtualization Servers

Best for: Full control, no recurring cloud costs after purchase

Tier	Hardware	Estimated Cost (R)	Strengths	Considerations
Entry-Level	Refurbished Mini PC (e.g., Lenovo ThinkCentre, Dell OptiPlex Micro) with Intel Core i5/i7, 32GB+ RAM, 512GB+ SSD	R3,000 - R8,000	Very low cost, low power consumption, physically small. A major upgrade from your current laptop.	Limited to 4-6 lightweight VMs. Not a true server; lacks redundancy features (RAID, dual power supplies).
Mid-Range	Refurbished Tower Server (e.g., Dell PowerEdge T-series) with 8+ core Xeon, 64GB+ RAM, RAID storage	R10,000 - R20,000	Good balance of power, expandability, and cost. Can run 8-15+ VMs. Provides server-grade reliability (ECC RAM, RAID for data protection).	Larger, noisier, and consumes more power than mini PCs.
High-Performance	New/Refurbished Rack Server (e.g., Dell PowerEdge R-series) with dual 16+ core Xeons, 128GB+ RAM, enterprise SSDs	R40,000 - R100,000+	High performance, maximum VM capacity (30+), excellent scalability, and enterprise-grade reliability.	High cost, high power consumption, noisy, requires a server rack or dedicated space.

Additional operating costs (monthly):

Expense	Entry-Level	Mid-Range	High-Performance
Electricity (est. R2.50/kWh)	R150-R250	R300-R500	R600-R1,200
Internet (static IP, business grade)	R800-R1,500	R800-R1,500	R1,50-R3,000
Cooling / space	Negligible	R200-R500	R1,000+
Total monthly OpEx	R1,000-R1,800	R1,300-R2,500	R3,100-R5,200

One-time CapEx: R3 000 - R100 000 depending on tier.

Advantages of refurbished virtualization servers:

- Lower long-term cost – one-time capital expense; no recurring cloud compute fees after purchase
- Full hardware control – no vendor restrictions on scraping behavior or data egress costs
- Performance isolation – no noisy neighbours; consistent CPU and I/O for 24/7 scraping
- High core density for the price – a refurbished server with 16+ cores costs less than equivalent cloud over 6–12 months
- RAID and ECC RAM (mid-range and higher) – hardware-level redundancy and error correction that a laptop or mini PC lacks
- Data sovereignty – all data stays on our premises; no third-party access concerns
- Predictable monthly OpEx – only electricity, internet, and cooling; no surprise API spikes

Disadvantages:

- High upfront capital required - R10,000 - R20,000 (mid-range) before we can start
- Physical space, cooling, and noise – servers need a secure, ventilated area; rack servers are loud
- Electricity and business-grade internet bills - adds R1,300 - R2,500 per month (mid-range)
- Hardware maintenance responsibility - failed power supply, disk, or fan may need to be replaced
- No built-in auto-scaling - limited to the physical cores in the server;
- Single point of physical location - if the space loses power or internet, all scraping stops
- Setup time of days to weeks - procuring, configuring, and provisioning takes significantly longer than cloud

Option B: Cloud-Native Alternative (No Physical Servers)

Best for: Speed to market, zero hardware management, built-in redundancy

Instead of buying servers, we run everything in the cloud with auto-scaling.

Service	Purpose	Recommended Provider	Monthly Cost (startup scale)
Scraping workers	Distributed scraping across retailers/locations	AWS EC2 Spot Instances or Render Background Workers	\$50-\$150 (~R900-R2,800)
Queue management	Coordinate scraping jobs	AWS SQS or Redis (Render)	\$0 - \$20
Database	Product + price data	MongoDB Atlas or PostgreSQL (see Section 4)	\$60 - \$250

Orchestration	Schedule & monitor scrapers	AWS Lambda (serverless) or Cron on Render	\$0 - \$30
Object storage	Backups, raw data exports	AWS S3 or Cloudflare R2	\$5 - \$20
Total monthly			~\$115 - \$470 (~R2,100-R8,700)

Scraping multiple locations: Spin up 10 - 20 concurrent cloud workers, each scraping a different suburb/postal code. This is impossible with a single laptop.

Advantages over physical servers:

- No upfront hardware cost
- Automatic failover (if one instance dies, another takes over)
- Pay only for what you use
- No electricity or internet bills to manage
- Can scale from 2 to 200 concurrent scrapers in minutes

Disadvantages:

- Recurring monthly costs (never own the hardware)
- Requires cloud expertise (or a part-time DevOps hire)

Section 4: Database Migration Options

SQLite is not production-ready for a growing startup. Below are three alternatives with pricing for our expected scale (assume 10,000 products × 100 price points each = 1M records).

Option 1: MongoDB Atlas (Recommended for price data)

Why MongoDB?	Schema flexibility (prices, retailers, locations vary widely); excellent for time-series price history
Plan	Shared cluster (M0 free tier) or M10 dedicated
Free Tier	512 MB storage, shared RAM - \$0 (good for prototyping)
M10 (starting production)	2 GB RAM, 10 GB storage - 0.08/hr (\$60/month or R1,100)
M20 (growth)	4 GB RAM, 20 GB storage - 0.17/hr (\$125/month or R2,300)
Backup (optional)	Point-in-time recovery - additional 0.05/hr(\$36/month)
Estimated monthly:	\$60 - \$160 (R1 100 - R3 000)

Option 2: PostgreSQL (Supabase or Neon)

Why PostgreSQL?	Strong data integrity, great for relational queries (products ↔ prices ↔ stores)
Supabase free tier	500 MB database, 2 GB transfer - \$0
Supabase Pro	8 GB database, 250 GB transfer - \$25/month (~R460)
Neon (serverless Postgres)	Free tier: 0.5 GB. Scale: 0.016/hrcompute+0.20/GB storage
Neon production estimate	~\$40 - \$80/month for your scale

Option 3: SQLite Cloud (Keep SQLite but make it production-ready) - Most Affordable

Service	Pricing	Features
	Free Tier: 1 GB, 3 million queries/month. Paid Tier: \$0.015/GB + \$0.50/million queries Our Estimated Usage (5GB,12Mqueries/month) = ~\$4.56/month (~R84) Estimated for growth (50 GB, 50M queries/month): ~\$24.24/month (~R448)	
Turso (SQLite edge database)		Distributed SQLite across edge locations, automatic backups, built-in replication, low-latency reads, supports full SQLite feature set
libSQL (open source)	Self-hosted: \$0	Same as Turso but you manage the infrastructure, no automatic backups, no edge distribution

Recommendation: For our current stage and immediate 6-12 months, Turso is likely the better choice. For long-term scaling to thousands of locations and real-time price history analytics, MongoDB may become preferable.

PocketBase (User Data)

PocketBase is fine for now. When user count exceeds 5,000, migrate authentication to Supabase Auth (free tier up to 50,000 users) or Auth0 (\$23/month for 1,000+ users).

Section 5: Platform Hosting Upgrade

Our current Render Starter plan (\$7/mo) is insufficient. Upgrade immediately to:

Plan	CPU	RAM	Monthly Cost	6-Month Cost
Standard (recommended now)	1 core	2 GB	\$25 (~R460)	R2 760
Professional (next 12 months)	2 cores	4 GB	\$85 (~R1416.57)	R8502
Auto-scaling (future)	Variable	Variable	\$100–\$300	N/A

Recommendation: Upgrade to Standard plan.

Section 6: Recommended 6-Month Roadmap

Month	Action	Est. Cost (USD)	Est. Cost (ZAR)	Notes
Month 1 (Weeks 1 - 2)	Launch cloud scraping bridge - AWS EC2 Spot or Render Background Workers. Add 3–5 locations immediately (JHB, Durban, Pretoria). Keep laptop as backup.	\$100 - \$150 (one-time)	R1,850 - R2,775	No waiting for Hardware. Location-based pricing within days.
Month 1 (Weeks 1–4)	Procure mid-range refurbished server (Dell PowerEdge T-series, 64 GB RAM, RAID). Begin delivery (1–3 weeks).	CapEx: ~\$1,080 (one-time)	R20,000	One-time hardware purchase.
Month 1 (Weeks 1–4)	Migrate database from local SQLite → Turso (free tier). Update connection string only.	\$0	R0	Automatic backups, no more single-file corruption.
Month 1 (Weeks 1–4)	Upgrade Render hosting to Standard plan (\$25/mo).	\$25/mo	R460/mo	Ongoing monthly cost
Month 2 (Weeks 5–6)	Server arrives & setup - Install Proxmox, configure VMs, networking, static IP, firewall.	\$0 (CapEx spent)	R0	

Month 2 (Weeks 5–6)	Deploy scrapers onto server VMs - 4-6 concurrent scrapers (retailers x locations).	\$0	R0	Server now handles core scraping load.
Month 2 (Weeks 5–6)	Set up proxy infrastructure - residential proxy pool (e.g. Smartproxy or Bright Data) to avoid IP bans as scraping scales across locations. See Section 8.2 for full options.	\$100-\$300/mo	R1,850- R5,550/mo	Required from this point onward. Without it, scrapers will be blocked within days of increased activity.
Month 2 (Weeks 5–6)	Begin phasing down cloud bridge - keep cloud only for excess locations or burst needs.	~\$50 (reduced)	~R925	Cloud costs drop significantly.
Month 3	Fully migrate all scraping to server - run 8-10 concurrent VMs. Add 2–3 more retailers and 5+ total locations. Terminate cloud instances (keep monitoring).	\$0 (server only)	R0	No more cloud bridge costs.
Month 3	Upgrade Turso to paid tier if exceeding free limits (>1 GB storage or >3 M queries/mo).	\$5 - \$10/mo	R90 - R185	Ongoing monthly
Month 4	Begin Scripting Automation + AI agent development (in-house) - auto-heal scrapers when retailer HTML changes. 85–130 hrs of engineering time across 4 phases. See Section 8 for full scope.	LLM API: \$30-\$100/mo (no dev cost - in-house)	R555-R1,850/mo (no cash dev cost)	In-house build — no contractor fees. Only ongoing cost is LLM API usage once each phase is deployed.
Month 5	Add off-site backups - nightly export from Turso to Cloudflare R2.	\$5 - \$10	R90 - R185	Protects against physical disaster (fire, theft, power surge).
Month 5	Set up monitoring & alerting - UptimeRobot, Sentry, or custom watchdog.	\$10 - \$20	R185 - R370	Immediate notification if server or scraper fails.
Month 6	Evaluate & optimise - server capacity, Turso performance.			Decision point: stay on Turso, add second server, or begin MongoDB migration.

Summary of 6-Month Costs

Category	Total (USD)	Total (ZAR)
One-time server CapEx	\$1,080	R20,000
Cloud bridge (Months 1 - 2, phased)	\$150 - \$200	R2,775 - R3,700
Render hosting (6 months × \$25)	\$150	R2,760
Turso database (Months 3 - 6: ~\$7.50/mo avg)	\$30 - \$45	R555 - R830
Proxy infrastructure (Months 2 - 6)	\$500 - \$1,500	R9,250 - R27,750
AI agent - LLM API only, in-house build (Months 4 - 6)	\$90 - \$300	R1,665 - R5,550
Off-site backups + monitoring (Months 5 - 6)	\$30 - \$60	R555 - R1,110
Monitoring & alerting (Month 5)	\$10 - \$20	R185 - R370
Total (excl. server CapEx)	\$935 - \$2,225	R17,298 - R41,163
Total (incl. server CapEx)	\$2,015 - \$3,305	R37,278 - R61,143

Note: The AI agent will be built in-house. No development or contractor fees apply - the only cash cost is LLM API usage once each phase is deployed. Engineering time (85 -130 hrs) should be planned into the team roadmap from Month 3 onward. See Section 8 for the full phase breakdown and skill requirements.

Section 7: 6-Month Infrastructure Budget

We estimate the capital expenditure for the next 6 months to be around R40 000 – R70 000. The majority of funds will go toward refurbished virtualization servers to eliminate our single point of failure, enable location-based pricing and increase data collection capability.

7.1 Priority 1: Refurbished Virtualization Servers (Scraping Infrastructure)

Tier	Hardware Example	One-Time CapEx (ZAR)	Why This Matters
Entry (Mini PC)	Lenovo ThinkCentre, 32 GB RAM	R3,000 - R8,000	Better than laptop, but limited to 4–6 VMs. No RAID.
Recommended (Mid-Range)	Dell PowerEdge T-series, 8+ core Xeon, 64 GB RAM, RAID	R10,000 - R20,000	Runs 8–15 VMs. Scrape 10+ retailers and 5+ locations concurrently. Hardware

			redundancy.
High-Performance (Future)	Dell R-series rack, dual Xeon, 128 GB RAM	R40,000 - R100,000+	For 30+ VMs. Not needed in next 6 months.

Allocated: R10,000 - R20,000 (one-time CapEx) - mid-range server.

Ongoing monthly OpEx (electricity, internet, cooling): approximately R1,300 - R2,500/mo. See Section 3 (Option A).

7.2 Priority 2: Cloud Bridge (Temporary - Months 1- 2 Only)

The cloud scraping bridge (AWS EC2 Spot or Render Background Workers) is not a 6-month recurring cost. It is a short-term measure used while the server is procured and configured. It is phased down in Month 2 and terminated entirely in Month 3.

Period	Purpose	Estimated Cost (USD)	Estimated Cost (ZAR)
Month 1	Launch 3-5 locations immediately; bridge while server is in transit	\$100 - \$150	R1,850 - R2,775
Month 2	Reduced - server takes over; cloud kept for burst/overflow only	~\$50	~R925
Month 3+	Terminated (monitoring only)	\$0	R0

Allocated: ~\$200 total (R3,700) - not a recurring monthly line item.

7.3 Priority 3: Data Storage & Database Migration

The recommended path is Turso for the immediate 6-12 months, with a future evaluation point for MongoDB at Month 6.

Solution	6-Month Cost (USD)	6-Month Cost (ZAR)	Notes
Turso (recommended)	\$0 - \$45	R0 - R830	Free tier covers early usage. Paid tier (~\$7.50/mo) kicks in only when exceeding 1 GB or 3 M queries/mo.

MongoDB Atlas M10 (future option)	\$360	R6,660	Considered at Month 6 evaluation. Not required in the initial 6-month window.
PostgreSQL / Supabase Pro	\$150	R2,775	Alternative if relational queries are preferred.

Allocated: ~\$30 - \$45 (R555 - R830) for 6 months on Turso.

7.4 Priority 4: Platform Hosting (Render Upgrade)

Plan	Monthly	6-Month (USD)	6-Month (ZAR)
Standard (1 CPU, 2 GB RAM) - recommended now	\$25	\$150	R2,760
Professional (2 CPU, 4 GB RAM) - upgrade when ready	\$85	\$510	R9,450

Allocated: \$150 (R2,760) for 6 months on Standard plan. Professional plan budgeted separately if upgrade is triggered mid-period.

7.5 Priority 5: Automation & AI Agent

Scripted automation is near-zero and implementable immediately. The AI agent is a development project and is fully scoped in Section 8.

Item	Type	Cost	Notes
Cron jobs, ETL scripts, watchdog daemons	Scripted - one-time setup	\$0 - \$100 (dev time)	Open-source tools. Can be implemented by existing team.
AI agent (auto-healing scrapers)	Development project + ongoing ops	See Section 8	Not a subscription - requires scoping, building, and testing before it generates any operational savings.
Monitoring & alerting (UptimeRobot, Sentry)	Ongoing	\$10 - \$20/mo	Know immediately if server or scraper fails.

Allocated for scripted automation + monitoring: \$0 - \$220 (R0 - R4,070) over 6 months. AI agent budget: see Section 8.

7.6 Priority 6: Contingency & Miscellaneous

Item	6-Month Cost (USD)	6-Month Cost (ZAR)
Off-site backups (Cloudflare R2 - from Month 5)	\$30 - \$60	R555 - R1,110
Unexpected cloud overages or hardware replacement parts	\$200 - \$400	R3,700 - R7,400

Allocated: ~\$300 (R5,550) contingency.

7.7 Revised 6-Month Investment Summary

Priority	Category	Cost Type	6-Month Cost (USD)	6-Month Cost (ZAR)
1	Refurbished server	One-time CapEx	\$1,080	R20,000
2	Cloud bridge (Months 1 - 2 only)	Temporary / phased out	\$200	R3,700
3	Database - Turso	Ongoing (from Month 3)	\$30 - \$45	R555 - R830
4	Render hosting - Standard plan	Ongoing monthly	\$150	R2,760
5a	Scripted automation + monitoring	One-time setup + small ongoing	\$100 - \$220	R1,850 - R4,070
5b	AI agent (built in-house) - LLM API + ops, Months 4-6	Ongoing ops only - no contractor cost	\$90 - \$300	R1,665 - R5,550
6	Contingency & off-site backups	Reserve	\$300	R5,550
	Total (excl. proxy infrastructure)		\$1,950 - \$2,295	R36,075 - R42,458
	Total (incl. proxy Infrastructure - See Section 8)		\$2,450 - \$3,795	R45,325 - R70,208

Note: AI agent development is handled in-house, so no contractor or external development cost is included. The only AI agent costs from Month 4 onward are LLM API usage and optional managed scraping services. Section 8 details the development phases and what the in-house team will need to build.

Section 8: Scraping Legality, IP Infrastructure & AI Agent Development

8.1 Legal & Compliance Considerations for Web Scraping in South Africa

Scraping retail websites carries legal and operational risk that is absent from most infrastructure discussions but is potentially the highest-impact risk to the business. This section addresses it honestly.

8.1.1 Retailer Terms of Service

South African retailers including Checkers, Pick n Pay, Woolworths, and Shoprite explicitly prohibit automated access in their terms of service. While civil enforcement against a small startup scraping publicly visible prices is uncommon, it is not unheard of, and the risk increases substantially as you scale from 2 to 10+ concurrent scrapers across multiple locations.

■ Legal Risk: Retailer ToS Violations

Automated scraping of retail sites is prohibited under most South African retailer terms of service.

IP bans and legal cease-and-desist letters become more likely at scale.

Mitigation options: rate limiting scrapers to human-like speeds, rotating IPs (see 8.2), and seeking data partnership agreements with retailers - which also produces a superior data product

8.2 IP Infrastructure: Proxies & Residential IPs

As scraping volume increases, retailer anti-bot systems will block the server's static IP address. A residential proxy network is the most effective countermeasure. This is a meaningful ongoing cost that must be budgeted.

Solution	How it Works	Est. Monthly Cost	Best For
Datacenter proxies	Shared IPs from hosting providers - cheap but easily detected and blocked	\$10 - \$50	Low-sensitivity sites, initial testing only
Residential proxies (e.g. Bright Data, Oxylabs, Smartproxy)	Real ISP-assigned IPs from consumer devices - very hard to detect	\$80 - \$300+ (usage-based, per GB)	Production scraping of major retailers
Mobile proxies (e.g. Bright Data Mobile)	4G/LTE mobile IPs - highest trust, lowest block rate	\$150 - \$500+	High-security sites that block datacenter and residential IPs
Rotating your own IPs (multiple SIM cards)	Manual IP rotation using multiple connections	R500 - R1,500/mo (SIM costs) + setup time	Cost-effective at small scale, difficult to maintain at 10+ locations

/ cloud regions)			
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Recommended approach: Start with Smartproxy or Bright Data's residential proxy pool at approximately \$100 - \$200/month for the expected scraping volume across 5 -10 locations. This cost is not optional at scale - without it, scrapers will be blocked within days of increased activity.

■ **Proxy Infrastructure: Required Cost from Month 2**

Residential proxy costs of \$100 - \$300/month are included in Sections 6 and 7 as a required line item from Month 2 onward.

Over 6 months (Months 2 - 6): approximately \$500 - \$1,500 total (~R9,250 - R27,750).

Without proxy infrastructure, scrapers will be blocked within days of scaling beyond a single static IP.

8.3 AI Agent Development: Scope & Effort (In-House)

The roadmap referenced an AI agent for auto-healing scrapers when retailer HTML changes. This will be built in-house, so no contractor or external development fees apply. However, it is still a meaningful engineering project with distinct phases and a non-trivial time investment from the team. Understanding the scope upfront prevents it from being treated as a quick weekend task.

8.3.1 What the Agent Actually Needs to Do

- Monitor scraper runs for failures and classify failure types (network, block, HTML change, rate limit)
- When an HTML change is detected, identify which CSS/XPath selectors are broken
- Use an LLM (e.g. Claude or GPT-4) to propose updated selectors based on the new HTML structure
- Validate the proposed selectors against a test page before applying them to production
- Alert a human if confidence is below threshold rather than applying a bad fix
- Log all changes with before/after diffs for auditability

8.3.2 Development Phases & Effort Estimate (In-House)

Phase	Description	Estimated Effort	Key Skills Required
Phase 1: Failure detection	Classify scraper failures by type; alert on HTML changes vs blocks vs network errors	20 - 30 hrs	Python, logging, basic ML classification

Phase 2: LLM selector repair	Prompt engineering + LangChain agent to propose new selectors from HTML diff	30 - 50 hrs	LangChain / LLM APIs, HTML parsing, prompt engineering
Phase 3: Validation & safety	Test proposed selectors before applying; confidence thresholds; human escalation	20 - 30 hrs	Testing frameworks, scraper integration
Phase 4: Deployment & monitoring	Integrate with existing scraper infrastructure; logging; dashboards	15 - 20 hrs	DevOps, systemd/Proxmox, dashboard tooling
Total		85 - 130 hrs	Buildable by one developer over 6–10 weeks part-time

At 85 - 130 hours of in-house development time, this represents roughly 6 - 10 weeks of part-time effort (10 - 15 hrs/week) or 3 - 5 weeks if prioritised as a primary focus. No external cost is incurred. Phase 1 alone (failure detection and alerting) can be completed in under 30 hours and delivers immediate operational value even before the full LLM-based selector repair is built.

8.3.3 Ongoing Operational Costs (Post-Build)

Once built, the AI agent's only hard costs are third-party API and service fees. In-house maintenance (occasional prompt tuning, selector logic adjustments) is absorbed by the team at no additional cash cost.

Item	Monthly Cost	Notes
LLM API usage (Claude / GPT-4)	\$30 - \$100	Depends on how frequently selectors break and how many retailers are monitored
Browse.ai or BrightData AutoScraper (optional)	\$50 - \$200	Managed selector services can reduce build complexity but add a recurring fee - evaluate after Phase 2

Total ongoing once built: \$30 - \$300/month (in-house) vs \$180 - \$600/month if a contractor were retained for maintenance. In-house development eliminates the largest cost component.

8.3.4 Revised Month 4 Budget Line for AI Agent

With in-house development, the Month 4 AI agent line reduces to operational costs only:

Item	Cost (USD)	Cost (ZAR)	Timing
Development (Phase 1–4) in-house	\$0 (time cost only)	R0	Month 4, 85 - 130 hrs of in-house engineering time
LLM API usage (Claude / GPT-4)	\$30 - \$100/mo	R555 - R1,850/mo	From Month 4 onward as each phase is deployed
Optional managed scraping service	\$50 - \$200/mo	R925 - R3,700/mo	Evaluate after Phase 2; not required from the start

■ Time Cost vs Cash Cost

In-house development eliminates the \$4,250 - \$10,400 contractor fee but the engineering time is a real opportunity cost. 85 - 130 hours spent on the agent is 85 - 130 hours not spent on product features, retailer integrations, or user growth. Phase 1 (failure detection) is recommended as a starting point - it delivers the highest value for the lowest time investment.

8.4 Revised Total Budget Including Proxy Infrastructure

Category	6-Month Total (USD)	6-Month Total (ZAR)	Notes
Refurbished server (CapEx)	\$1,080	R20,000	One-time
Cloud bridge (Months 1-2)	\$200	R3,700	Temporary
Render hosting (Standard, 6 mo)	\$150	R2,760	Ongoing
Turso database	\$30 - \$45	R555 - R830	From Month 3
Scripted automation + monitoring	\$100 - \$220	R1,850 - R4,070	One-time setup
Proxy infrastructure (Months 2 - 6)	\$500 - \$1,500	R9,250 - R27,750	Required at scale - residential IPs
AI agent - LLM API only (Months 4-6, in-house built)	\$90 - \$300	R1,665 - R5,550	No dev cost; LLM API only
Contingency	\$300	R5,550	Reserve
Total	\$2,450 - \$3,795	R45,330 - R70,210	

